

VIVEK V. VEKARIYA

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EDUCATION

2023 – Present

Doctor of Philosophy
Technical University of Munich, Germany

2017 – 2020

Master of Science in Information Technology (Embedded Systems)
University of Stuttgart, Germany

CGPA : 2.2

Related Courses

- Deep learning, Detection and Pattern Recognition (ML), Matrix computations for Signal Processing and Machine Learning.

2011 – 2015

Bachelor of Engineering in Instrumentation and Control
Gujarat Technological University, India

CGPA : 1.38 (Course Topper)

Related Courses

- Control Theory, Microprocessors and Microcontrollers, Distributed Control Systems (DCS).

ACADEMIC PROJECTS

2019 – 2020

Master Thesis: Evaluating Robustness of Stochastic Neural Networks against Adversarial Learning Attacks

- Implemented Stochastic Neural Nets (SNN) for hardware efficient applications eg. IoT & wearable devices.
- Evaluated robustness of SNN and CNN against Carlini and Wagner (C&W) & One-pixel attack methods.
- Found that SNNs are more robust than CNNs against adversarial attacks under gray-box & black-box settings.

2019

Conditional Convolutional Neural Network for Medical Image Segmentation

- Applied **Dynamic Filter Network** on U-Net architecture to specify conditional inputs for semantic segmentation on 3D MRI data.

2018

Enhancing user interaction with the smartphone using Phalanx bone as an additional input

- Collected and preprocessed data through user study using Android application.
- Performed blob detection and data augmentation using OpenCV, Pandas and Numpy.
- Implemented **VGG based CNN** architecture for classification of finger and phalanx touch and achieved **98.05%** accuracy.

WORK EXPERIENCE

Apr 2024 – Present	Scientific Researcher	Technical University of Munich
	- Development and research on a comprehensive framework for dataset quality management in Deep Learning applications, encompassing test dataset assessment, generation, prioritization, and robustness enhancement. - Initiated and managed practicum and seminar series at TUM focused on the current AI Testing landscape.	
Mar 2021 – Mar 2024	Scientific Researcher	fortiss GmbH
	- Developed metrics to quantify the dataset quality for DL applications based on latent space of Deep Neural Networks. - Developed a framework to sample corner cases for the Pedestrian Detection System during the project KI-Absicherung (AI-Validation) based on coverage-guided fuzzing. - Trained and Tested Single Shot Multi-box Detector (SSD) for 2D Object Detection using Automated Driving datasets.	
Apr 2020 – Jun 2020	Intern in Artificial Intelligence	FARO Europe GmbH & Co. KG
	- Generating 3D model from 2D images using AI which included research about various approaches, generating training dataset, designing NN using LSTM / GRU blocks etc.	
Oct 2019 – Mar 2020	Work Student in Artificial Intelligence	FARO Europe GmbH & Co. KG
	- Worked on Computed Tomography (CT) voxel data for high dimensional 2D image pre-processing and reconstruction using Python libraries. - Designed and trained Auto-Encoders and CNNs (Tensorflow & Keras) for industrial CT machines.	
Dec 2018 – Jul 2019	Work Student	Robert Bosch GmbH
	- Developed business KPI dashboards using Python for visualizing daily progress of software development life cycle.	
Mar 2018 – Aug 2018	Work Student	Harman Becker Automotive Systems GmbH
	- Developed a splitting tool for log and trace files for navigation systems for Daimler.	
Jul 2015 – Aug 2017	Executive Project Engineer	Yokogawa India Ltd.
	- Worked on System Engineering of Distributed Control System (DCS) for green field projects in Oil & Gas Industries. - Involved in factory acceptance testing at Abu-Dhabi and Dubai locations.	

TECHNICAL SKILLS

Languages	Python, C, C++, SQL.
DL Frameworks	TensorFlow, Keras, PyTorch.
Python Libraries	OpenCV, NumPy, Pandas, Scikit-learn, Matplotlib, PyQT.
DevOps Tools	Docker, Bitbucket, Git.
OS	Windows, Linux.
Others	Power BI, Tableau, MATLAB, LabVIEW, Microsoft Office, Jira, Scrum.

CERTIFICATIONS

- Structuring Machine Learning Projects, Sequence Model, Improving Deep Neural Networks: Hyper-parameter tuning, Regularization and Optimization (Coursera).
- SQL for Data Science (Coursera).

LANGUAGES

English	Business Fluency
German	Advanced Proficiency (B1)
Gujarati, Hindi	Native/Bilingual

Publications

- (1) "Latent Space Class Dispersion: Effective Test Data Quality Assessment for DNNs"
Vivek Vekariya, Mojdeh Golagha, Andrea Stocco, Alexander Pretschner, Full Paper Access:
<https://arxiv.org/abs/2503.18799>
- (2) "Stochastic computing as a defense against adversarial attacks"
Florian Neugebauer, Vivek Vekariya, Ilia Polian, John P Hayes, Full Paper Access:
<https://ieeexplore.ieee.org/abstract/document/10207101>
- (3) "Segmentation of Computed Tomography Voxel Data using Machine Learning"
Ariane Stiebeiner, Georgios Balatzis, Vivek Vekariya, USA Patent Number: 20230394718